CHAPTER 21 — AIR DISTRIBUTION (VENTILATION)

CONTENTS — MAINTENANCE PROCEDURES

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AIR DISTRIBUTION (VENTILATION)

HEATING AND VENTILATION SYSTEM

21-1. HEATING AND VENTILATION AIR DISTRIBUTION SYSTEM.

21-2. DESCRIPTION.

The cabin air distribution system (figure 21-1) includes two subsystems which provide optional use of heated or ventilated air from separate sources.

Heated air is supplied to lower window defroster nozzles and to four windshield defroster nozzles for cabin heating-defrosting using bleed air from engine compressors as heat source.

Ventilating system uses ram air as source from ventilating air intakes on cabin nose with optional use of blowers for ventilating air outlets ahead of pilot-copilot stations (figure 21-2).

21-3. VENTILATION SYSTEM.

Ventilation system (figure 21-2) delivers outside air to ventilating air outlets ahead of pilot-copilot stations. Ram air inlet grills are located on cabin nose below each windshield. Ducts extending from air inlet grills direct ram air to ventilating air outlets. Air intake volume is controlled by push/pull knobs near ventilating air outlets. When ram air is low, outside air may be pulled in by turning "ON" the blower which is part of the system.

21-4. VENTILATION COMPONENTS.

21-5. Removal.

NOTE

The removal procedure for left plenum and blower is given. The removal procedure for the right plenum and blower is similiar.

1. Remove ten screws (1, figure 21-3) and remove ram air inlet grill (2) on left side of cabin nose.

2. Remove tie wraps (21) from tubes (20, 24, and 25).

3. Remove three clamps (19).

4. Remove tubes (20, 22, 24, and 25) and adapter (23) between grommet (18) and bottom of plenum (26).

5. Remove clamp (9) and outlet plenum (5).

6. Remove clamp (12) and remove plenum (26) as a unit.

7. Remove clamps (15 and 17) and remove ventilation blower (16).

21-6. Installation.

MATERIALS REQUIRED

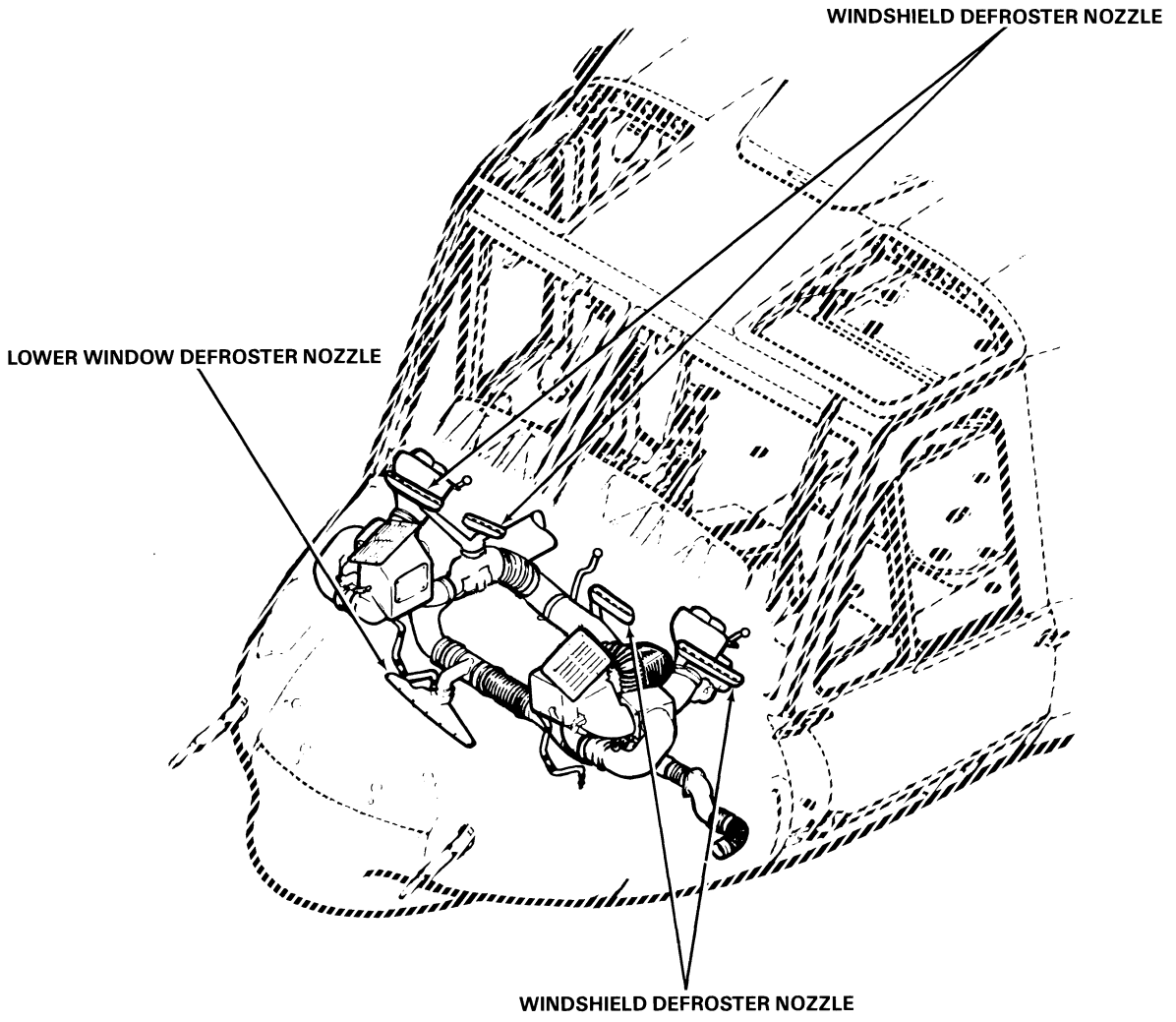
NUMBER	NOMENCLATURE
MS3367-5-9	Wraps, Tie

NOTE

The installation procedure for left plenum and blower is given. The installation procedure for the right plenum and blower is similar.

1. Position plenum (26, figure 21-3) in cabin nose and install clamp (12), screw (10), washer (11), and spacer (13). Thread screw (10) into threaded insert on electrical equipment shelf.

2. Position clamp (9) on outlet plenum (5) and plenum (26). Install screw (8), washer (7), and nut (6).



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Figure 21-1. Heating and ventilation air distribution system

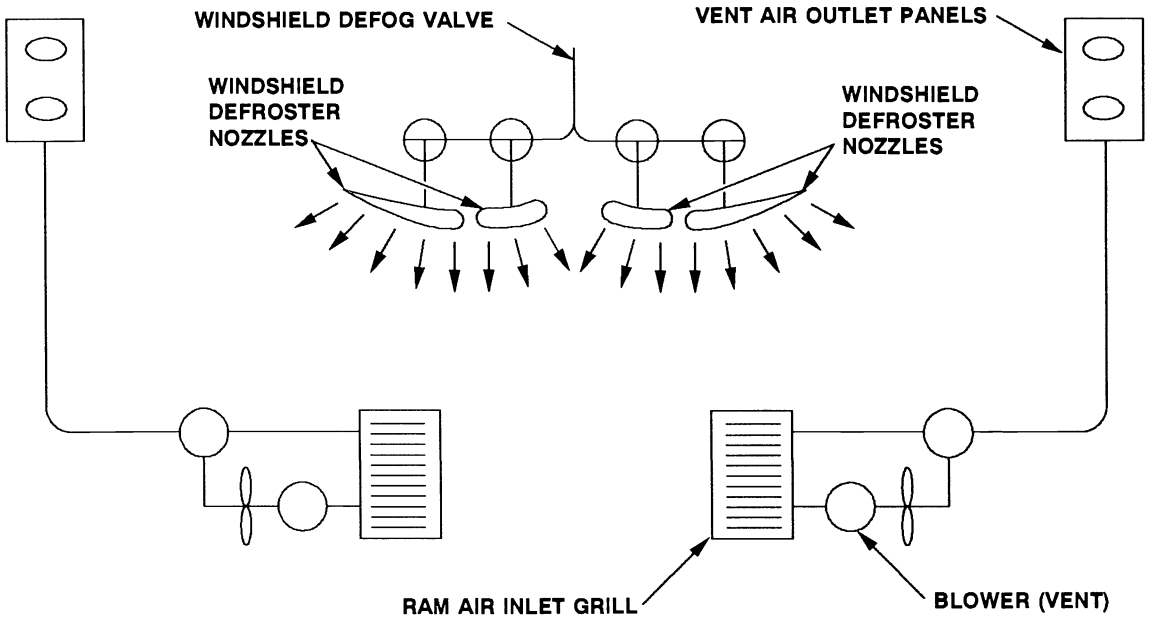


Figure 21-2. Heating and ventilation air distribution system schematic

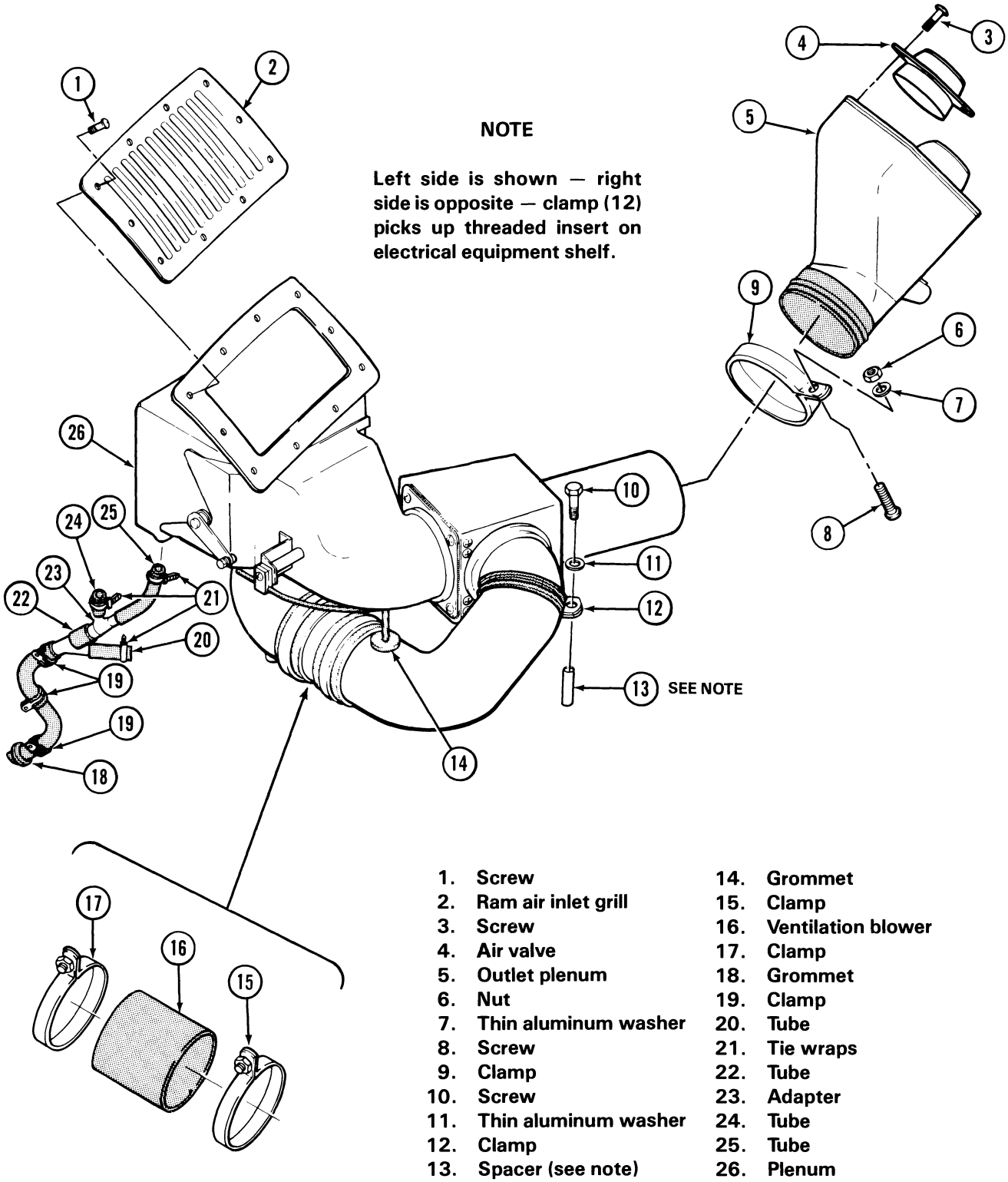
3. If not previously accomplished, position air valve (4) on outlet plenum (5) and install four screws (3).

4. Position tubes (25, 24, 22, and 20) and adapter (23) on plenum (26). Install three clamps (19) and install tiedown wraps (21) on tubes (25, 24, 22, and 20).

5. If not previously accomplished, position ventilation blower (16) on plenum (26), and install clamps (15 and 17).

6. Position ram air inlet grill (2) on top of cabin nose left side. Install ten screws (1) to secure grill to plenum (26).

7. Perform functional check of ventilation blower (16).



1. Screw
2. Ram air inlet grill
3. Screw
4. Air valve
5. Outlet plenum
6. Nut
7. Thin aluminum washer
8. Screw
9. Clamp
10. Screw
11. Thin aluminum washer
12. Clamp
13. Spacer (see note)
14. Grommet
15. Clamp
16. Ventilation blower
17. Clamp
18. Grommet
19. Clamp
20. Tube
21. Tie wraps
22. Tube
23. Adapter
24. Tube
25. Tube
26. Plenum

Figure 21-3. Ventilation components installation